

$$f = 415 \text{ N}$$

[A1]

2 (a)

Newton's law of gravitation states that the attractive gravitational force between two point masses [A1] is directly proportional to the product of the masses and inversely proportional to the square of their separation [M1].

2 (b)

The gravitational force acting on the moon provides the centripetal force. [B1]

2018 JC2 H2 Physics Prelims Exam Solution (9749 Paper 3)

$$F_c = ma_c$$

$$\frac{GMm}{x^2} = m\omega^2 x \quad [M1]$$

$$\text{Since } \omega = \frac{2\pi}{T}$$

$$\frac{GMm}{x^2} = m\omega^2 x \left(\frac{2\pi}{T} \right)^2 \quad [M1]$$

$$GM = \frac{4\pi^2 x^3}{T^2} \quad [A0]$$

2(c)(i)

$$\text{gradient} = \frac{(4.5 \times 10^{14} - 1.3 \times 10^{14}) \times (1000)^3}{(0.35 - 0.10) \times (24 \times 60 \times 60)^2} \quad [M1]$$

$$= 1.71 \times 10^{14} \text{ m}^3 \text{ s}^{-2} \quad [A0]$$

2(c)(ii)

$$GM = \frac{4\pi^2 x^3}{T^2}$$

$$6.67 \times 10^{-11} \times M = 4\pi^2 \times 1.71 \times 10^{14} \quad [C1]$$

$$M = 1.02 \times 10^{26} \text{ kg} \quad [A1]$$

2(d)

The gradient of the graph is equal to $\frac{GM}{4\pi^2}$. Since the mass of planet Uranus is smaller than that of planet Neptune, the graph for Uranus is one with smaller gradient [B1] with the same zero-y-intercept [B1].

2(e)